

DAX Assignment – Top 20 Questions

Employee Basic Calculations

Employee Age = INT(DATEDIFF(emp[DOB], TODAY(), DAY) / 365.25)

1. Create a **calculated column** to display **Employee Age** using DOB.
 2. Create a **measure** to calculate **average age of employees**.
 3. Create a **calculated column** to find **years of experience** using DOJ.
 4. Create a **measure** to calculate **average experience** of all employees.
 5. Create a **measure** to count **total number of employees**.
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Date-Based Analysis

6. Create a measure to count employees **joined this year**.
 7. Create a measure to count employees **joined in the last 5 years**.
 8. Create a calculated column to extract **Joining Year** from DOJ.
 9. Create a calculated column to extract **Birth Month** from DOB.
 10. Create a measure to count employees **born in each month**.
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Filtering & Conditional Logic

11. Create a calculated column to classify employees as **“Senior” (Experience ≥ 10 years)** or **“Junior”**.
 12. Create a measure to count **Senior employees**.
 13. Create a measure to count employees **above 40 years of age**.
 14. Create a calculated column to check whether **Email ID is valid** (contains "@").
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Text & Data Quality

15. Create a calculated column to **mask phone numbers** (show last 4 digits only).
 16. Create a calculated column to display **Full Employee Info** (NAME + PHONE).
 17. Create a measure to count employees with **missing Email ID**.
 18. Create a calculated column to identify **duplicate phone numbers**.
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Advanced / Analytical

19. Create a measure to calculate **employee attrition risk** (Employees with age > 50 and experience > 25 years).
20. Create a ranking measure to **rank employees by experience**.

DAX Assignment – Solutions (Top 20)

Employee Basic Calculations

1. Employee Age (Calculated Column)

Employee Age =

```
DATEDIFF(Employee[DOB], TODAY(), YEAR)
```

2. Average Age of Employees (Measure)

Average Age =

```
AVERAGEX(  
    Employee,  
    DATEDIFF(Employee[DOB], TODAY(), YEAR)  
)
```

3. Years of Experience (Calculated Column)

Experience Years =

DATEDIFF(Employee[DOJ], TODAY(), YEAR)

4. Average Experience (Measure)

Average Experience =

AVERAGEX(
Employee,
DATEDIFF(Employee[DOJ], TODAY(), YEAR)
)

5. Total Number of Employees (Measure)

Total Employees =

COUNT(Employee[EID])

Date-Based Analysis

6. Employees Joined This Year (Measure)

Joined This Year =

CALCULATE(
COUNT(Employee[EID]),
YEAR(Employee[DOJ]) = YEAR(TODAY())
)

7. Employees Joined in Last 5 Years (Measure)

Joined Last 5 Years =

CALCULATE(
COUNT(Employee[EID]),

```
Employee[DOJ] >= DATE(YEAR(TODAY()) - 5, 1, 1)  
)
```

8. Joining Year (Calculated Column)

Joining Year =
YEAR(Employee[DOJ])

9. Birth Month (Calculated Column)

Birth Month =
FORMAT(Employee[DOB], "MMMM")

10. Employees Born Each Month (Measure)

Employees by Birth Month =
COUNT(Employee[EID])

Use Birth Month in Rows and this measure in Values

Filtering & Conditional Logic

11. Senior / Junior Classification (Calculated Column)

Employee Level =
IF(
 Employee[Experience Years] >= 10,
 "Senior",
 "Junior"
)

12. Count Senior Employees (Measure)

Senior Employees =

```
CALCULATE(  
    COUNT(Employee[EID]),  
    Employee[Experience Years] >= 10  
)
```

13. Employees Above 40 Years (Measure)

Employees Age Above 40 =

```
CALCULATE(  
    COUNT(Employee[EID]),  
    DATEDIFF(Employee[DOB], TODAY(), YEAR) > 40  
)
```

14. Valid Email Check (Calculated Column)

Valid Email =

```
IF(  
    CONTAINSSTRING(Employee[EMAIL], "@"),  
    "Valid",  
    "Invalid"  
)
```

Text & Data Quality

15. Mask Phone Number (Calculated Column)

Masked Phone =

```
"XXXXXX" & RIGHT(Employee[PHONE], 4)
```

16. Full Employee Info (Calculated Column)

Employee Info =

Employee[NAME] & " - " & Employee[PHONE]

17. Missing Email Count (Measure)

Missing Email Count =

```
CALCULATE(  
    COUNT(Employee[EID]),  
    ISBLANK(Employee[EMAIL])  
)
```

18. Duplicate Phone Numbers (Calculated Column)

Duplicate Phone =

```
IF(  
    CALCULATE(  
        COUNT(Employee[EID]),  
        ALLEXCEPT(Employee, Employee[PHONE])  
    ) > 1,  
    "Duplicate",  
    "Unique"  
)
```

Advanced / Analytical

19. Employee Attrition Risk (Measure)

Attrition Risk Employees =

```
CALCULATE(  
    COUNT(Employee[EID]),  
    DATEDIFF(Employee[DOB], TODAY(), YEAR) > 50,  
    DATEDIFF(Employee[DOJ], TODAY(), YEAR) > 25
```

)

20. Rank Employees by Experience (Measure)

Experience Rank =

RANKX(

ALL(Employee),

DATEDIFF(Employee[DOJ], TODAY(), YEAR),

,

DESC,

DENSE

)